Prolog - Loop & Decision Making

1. Loops

- Loop statements are used to execute the code block multiple times.
- Prolog has no looping facility, but we can obtain a similar effect. Using this effect, we can evaluate a sequence of goals repeatedly. In various ways, this can be done like built-in predicates, recursion, backtracking, or a combination of these.
- The following program outputs the integer value from a specified value down to 1.

Try the following

```
count_to_10(10) :- write(10),nl.
count_to_10(X) :-
   write(X),nl,
   Y is X + 1,
   count_to_10(Y).
```

- In the above example, we define the count_to_10 predicate in terms of itself.
- The **first clause** read as 'to **count_to_10** from 10.
- This process will terminate using the **first clause**: 'when the argument is 10, then stop or do nothing'.
- Here the first clause is called as **terminated condition**.

Query 1: count_to_10(3).

Output

```
3
4
5
6
7
8
9
10
true
```

Looping a fixed number of times

- The **testloop** predicate is defined as ‘loop from **N**, write the value of **N**, then subtract one to give to **M**, then loop from **M**’. ‘And by the first clause, is defined as ‘if the argument is **zero**, do nothing(**stop!**)’. test of the program.

Try the following

```
testloop(0)
testloop(N) :- N>0, write('Number : '), write(N), nl, M is N-1, testloop(M).
```

Query 1: testloop(3).

Output

```
Number : 3
Number : 2
Number : 1

true ?
```

Looping until a condition is satisfied

- This example below shows the use of recursion to read terms entered by the user from the keyboard and output them to the screen, until a **word end** is encountered, using ‘disjunctive goal’(**word=end**).

Try the following

```
test :- write(‘Type the word : ‘), read(word), write(‘Input was ‘), write(word), nl, (word=end; test).
```

Query 1:

```
?- test.
Type the word : Welcome.
Input was Welcome
Type the word : Tceusl.
Input was Tceusl
Type the word : end.
Input was end.
True
```

Decision Making

- The decision statements are **If-Then-Else** statements. So when we try to match some condition, and perform some task, then we use the decision making statements.
- The basic usage is as follows

  If <condition> is true, **Then** <do this>, **Else**

- In some different programming languages, there are If-Else statements, but in Prolog we have to define our statements in some other manner.

- Following is an example of decision making in Prolog.

```
% If-Then-Else statement

gt(X,Y) :- X >= Y,write('X is greater or equal').
gt(X,Y) :- X < Y,write('X is smaller').

% If-Else if-Else statement

gte(X,Y) :- X > Y,write('X is greater').
gte(X,Y) :- X =:= Y,write('X and Y are same').
gte(X,Y) :- X < Y,write('X is smaller').
```

- Type the above code and try the following Queries

Query 1: gt(10,100).

Output

X is smaller

Yes

Query 2: gt(150,100).

Output

X is greater or equal

true ?

Query 3: gte(10,20).

Output

X is smaller

Yes

Query 4: gte(100,100).

Output

X and Y are same

true ?

Review Exercises

Q1. Try the following Prolog program and try the sample query below.

```
count_down(L, H) :-
   between(L, H, Y),
   Z is H - Y,
   write(Z), nl.

count_up(L, H) :-
   between(L, H, Y),
   Z is L + Y,
   write(Z), nl.
```

Query 1: count_down(12,17).

Query 2: count_up(5,12).

What is your understanding about this output?

Q2. Define a predicate to output the values of the squares of the integers N1 to N2 inclusive and test it with N1=6 and N2=12.

Q3. Define a test predicate to read a serious of characters input by the user and output all of those before the first **newline** or **?** Character.

Q4. This example of program below is designed to search database containing clauses representing the name, age, places of residence, and occupation of a number of people.

The database contains these values :

```
person(john,smith,45,london,doctor).
person(martin,williams,33,birmingham,teacher).
person(henry,smith,26,manchester,plumber).
person(jane,wilson,62,london,teacher).
person(mary,smith,29,glasgow,surveyor).
```

Q4. Cont.

(a) Define a predicate **somepeople** below is to search the person in the database which surname is smith.

Sample Query:
?- somepeople.

Output
john smith
henry smith
mary smith
True

(b) Using the **person** clause given in **Q4** find the professions of all those over age 30.